Generalized GHS: checked descent and attack costs

Exact binary curves, subgroup-preserving maps, prior-work classification, and matched target costs | 25 October 2026

10	232	21	0
completed attack edges	equivalent genus-two profiles	zero-image profiles	new vulnerabilities established

Finding. In the four completed screens, every surviving low-genus Hess parameter reduces to one of four canonical quadratic covers. The executed curves lie in known GHS/isogeny families. The recorded one-sample costs do not support a calibrated advantage over rho.

Exact source curves

GF(2^n)	modulus	a, b	#E / r	P	Q
n=6	0x43	0, 32	76 / 19	(6, 31)	(6, 25)
n=8	0x11b	0, 32	268 / 67	(244, 61)	(4, 8)
n=10	0x409	0, 32	1032 / 43	(262, 371)	(626, 155)
n=12	0x1009	1, 512	3980 / 199	(23, 94)	(2212, 1499)

Each curve has equation `y^2 + x*y = x^3 + a*x^2 + b` in the stated polynomial-basis field. The source subgroup generator and target coordinates are part of the exact instance identity; the target scalar is 18 for n=6 and 37 for the other three curves.

Classification

The screen found 253 genus-at-most-two profiles. A Jacobian-order bound excludes 21 because the selected order-199 subgroup cannot have a nonzero image. The other 232 are parameter presentations of four direct covers already executed end to end. Ten direct or isogeny attack edges are complete. No exact instance is classified as a new vulnerable curve; absence of a publication match for its hexadecimal coefficients does not establish novelty.

Scope is deliberately exact: four source curves, their listed subfields and descent parameters, and six small rational-kernel isogeny edges. It is not a census of every binary elliptic curve or every isogeny chain.

Why 232 parameters collapse

Write `beta = sqrt(b)` and `d = beta/c`. The documented Artin-Schreier presentation is `v^2 + v = c/z + a + d*z`, with `c*d = beta`. For all 232 index-two genus-two profiles in these exact screens, exactly one of `c` and `d` lies in the subfield. If `d` does, put `w=d*z`; otherwise put `w=c/z`. Both give `v^2 + v = beta/w + a + w` by a subfield-rational change.

In the second branch the source point is presented as `T-P`, where `T=(0,beta)` is rational two-torsion. Subtracting the transformed origin makes the induced degree-zero transfer the negative of the canonical transfer, preserving the discrete-log relation.

n	subfield degree	canonical c	genus-two c values	subgroup points checked
6	3	0x24	14	18
8	4	0xc5	30	66
10	5	0x42	62	42
12	6	0x208	126	198

Map check. Each of the ten executed composite maps was checked on every scalar of its selected prime-order subgroup, with nonzero Jacobian image and preserved target relation. Sage independently checked the 232 parameter identities and all 324 nonzero subgroup-point identities. The normalized presentations were not separately timed as 232 Jacobian implementations.

Six rational-kernel isogeny edges

source n	degree	kernel generator	quotient (a, b)
6	2	(0, 36)	(0x0, 0x24)
8	2	(0, 197)	(0x0, 0xc5)
10	2	(0, 66)	(0x0, 0x42)
10	3	(861, 103)	(0x0, 0x15a)
12	2	(0, 520)	(0x1, 0x208)
12	5	(8, 576)	(0x1, 0x249)

The full curve-to-cover formulas, quotient maps, and exact reduced Mumford images of `P` and `Q` are preserved per edge in the raw JSON receipts and the detailed REPORT.md. Quotient equations were checked on all rational source points; the selected subgroup was checked exhaustively.

Measured GHS versus rho work

All elapsed values below are milliseconds from one unisolated sample per edge. Setup includes cover construction, source-order enumeration, and exhaustive subgroup-map verification; isogeny rows also include kernel and quotient validation. GHS precompute covers the factor base, relation collection, and matrix work. GHS online covers target mapping, descent, and replay. Rho is the complete negation-folded run on the same source `(P,Q)`.

n	edge	FB / rank	smooth / tries	setup	GHS pre	GHS on	rho on
6	direct	8 / 7	10 / 18	2.067	3.758	0.156	0.056
6	two-isog	8 / 7	10 / 18	10.516	1.192	0.157	0.021
8	direct	16 / 16	50 / 66	17.486	24.064	0.244	0.050
8	two-isog	16 / 16	50 / 66	18.431	11.754	0.897	0.026
10	direct	32 / 22	26 / 42	25.501	24.328	0.267	0.061
10	two-isog	32 / 22	26 / 42	27.865	10.188	1.613	0.128
10	three-isog	32 / 18	26 / 42	33.400	16.653	0.237	0.313
12	direct	64 / 56	132 / 198	74.308	144.882	0.253	0.190
12	two-isog	64 / 56	132 / 198	148.731	111.973	0.791	0.175
12	five-isog	64 / 56	132 / 198	185.539	108.527	0.353	0.176

Cost interpretation. The n=10 degree-3 edge has a lower recorded GHS online time (0.237 ms) than this rho run (0.313 ms), but it also incurs 33.400 ms of setup and 16.653 ms of GHS precomputation. These samples are too small and unisolated for a speedup claim.

What the counters include

The receipts separately tally finite-field additions, multiplications, squarings, and informational inverse calls; they also record matrix-entry operations, polynomial-root tests, hash work, and rho additions, doublings, steps, and restarts. Inverse-call tallies overlap with counted field operations and must not be added as an independent cost. Source-field rho, smaller-field Jacobian arithmetic, and matrix work use different units, so the elapsed columns are not an algorithm-independent operation ratio.

No calibrated GAE band or comparison against the fastest external rho implementation was completed. The report preserves the original single-sample receipts rather than reselecting favorable timings after the fact.

Proof and replay status

check	result	boundary
Lean axiom audit	13 declarations; 0 failing; 0 native_decide trusted	Abstract scalar transfer, push-pull injectivity, zero-image criterion, normalization, sign
Executable maps	10/10 full subgroup checks	Concrete trace-conorm and Velu formulas checked on these finite groups
Independent Sage	4 curve orders; 10 receipt replays; 232 c identities	324 nonzero selected subgroup-point identities
Rust focused tests	26 library + 4 attack executable passed	Integrated PR head 16f786d3d26

Lean does not yet prove the concrete trace-conorm or Velu coordinate maps, and the Weil bound arithmetic is audited outside Lean. The 21 excluded profiles have zero image for the selected order-199 subgroup by the stated bound; that result does not classify arbitrary subgroups on other curves.

Canonical direct-cover data

For the four direct edges, `C: W^2 + hW = f` with `h=[0x0,0x1,0x1]`. Coefficient arrays run from constant term upward. The basis embeds the smaller field in the source field; the last column gives the reduced Mumford images of the exact source generator and target.

n	f coefficients	subfield basis	J(P) / J(Q) as (u; v)
6	[0x0, 0x7, 0x7, 0x3, 0x0, 0x3]	[0x1, 0xe, 0x17]	P: u=[0x4, 0x0, 0x1]; v=[0x7, 0x4] Q: u=[0x4, 0x0, 0x1]; v=[0x3, 0x5]
8	[0x0, 0x5, 0x5, 0x9, 0x0, 0x9]	[0x1, 0xc, 0x50, 0xed]	P: u=[0xf, 0x1]; v=[0xb] Q: u=[0xa, 0x5, 0x1]; v=[0xd, 0x1]
10	[0x0, 0x6, 0x6, 0x1c, 0x0, 0x1c]	[0x1, 0xe, 0x54, 0x358, 0x134]	P: u=[0x7, 0x18, 0x1]; v=[0x0, 0x9] Q: u=[0xd, 0x18, 0x1]; v=[0x17, 0x13]
12	[0x0, 0x8, 0x9, 0x9, 0x1, 0x9]	[0x1, 0x20, 0x400, 0x48, 0x900, 0x82]	P: u=[0x7, 0x2, 0x1]; v=[0x3e, 0x19] Q: u=[0x38, 0x12, 0x1]; v=[0x18, 0x1c]

Prior work and candidate status

The descent model and its parameter c are documented by Magma's Weil Descent handbook. Moving to isogenous curves before GHS is covered by Galbraith-Hess-Smart (EUROCRYPT 2002); Hess (2004) generalizes the construction. The per-edge coefficient searches and exact-instance uncertainty are recorded in PRIOR_WORK.md. All ten edges are classified as known-family examples, with exact publication matches unresolved. No new vulnerability claim survives this exact set.

Reproduce and inspect

Research directory: `research/ghs_generalized_20261024/`. The detailed `REPORT.md` and `results/exact-*.json` contain every image and raw cost vector. `results/MANIFEST.json` binds the measured executable and receipt hashes. Run `verify_orders.sage`, `verify_receipts.sage`, and `verify_c_equivalence.sage` with Sage; run the Lean axiom audit from `lean/`. The review branch is PR #2539.

Research limit. This report covers the quadratic genus-two executable branch on four specified curves. It does not establish an exhaustive classification across all fields, higher-genus index calculus, long isogeny walks, or a calibrated advantage over rho.